

$$\textcircled{1} \quad a = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

Find P ? a through O

$$\left(\frac{a a^T}{a^T a} \right) b$$

$\textcircled{P}$

$$\Rightarrow \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \begin{bmatrix} 2 & 1 & 2 \end{bmatrix}$$

$$(4 + 1 + 4)$$

$$\Rightarrow \frac{1}{9} \begin{bmatrix} 4 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 4 \end{bmatrix}$$

b) What are eigenvalues of P

$$P \Rightarrow 0, 1, 0$$

$$0 + 1 + \lambda = 1$$

$$\boxed{\lambda = 0}$$

c) What are eigenvectors?

$$\Rightarrow \lambda = 1 \quad x_1 = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

$$\lambda_2 = 0 \quad x_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} ?$$

$$\lambda_3 = 0 \quad x_3 = \begin{bmatrix} 1/2 \\ -i \\ 0 \end{bmatrix} ?$$

$$d) \quad u_{k+1} = P u_k \quad u_0 = \begin{bmatrix} 9 \\ 9 \\ 0 \end{bmatrix}$$

$$u_{k+1} = P^k u_0 = P u_0$$

$$u_1 = \begin{bmatrix} 4 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 4 \end{bmatrix} \begin{bmatrix} 9 \\ 9 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 6 \end{bmatrix}$$

$$u_1 = a \frac{a^T u_0}{a^T a} = \underline{\underline{3a}}$$

$$u_2 = \frac{a a^T (3a)}{a^T a} = \underline{\underline{3a}} \left(\frac{a^T a}{a^T a} \right)$$

$$u_2 = 3a$$

$$u_3 = 3a$$

Because P is projection matrix

$$P^k = P \quad A^k u_0 = c_1 \lambda_1^k x_1 + \dots$$

②

$$t = 1$$

$$2$$

$$3$$

$$y = 4$$

$$5$$

$$8$$

$$\Rightarrow \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 8 \end{bmatrix}$$

$$y = Dt$$

$\Rightarrow$

$$1 \cdot D = 4$$

$$2 \cdot D = 5$$

$$3 \cdot D = 8$$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} D = \begin{bmatrix} 4 \\ 5 \\ 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \hat{D} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 8 \end{bmatrix}$$
$$14 \hat{D} = 38$$
$$\hat{D} = \frac{38}{14} = \frac{19}{7}$$

b) What vector is projected
on what line?

$$\begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ 8 \end{bmatrix}$$

onto

$$\begin{bmatrix} 38/14 \\ 1 \end{bmatrix}$$

$$\textcircled{3} \quad a_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad a_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Find 2 orthogonal vectors in this plane.

$$\Rightarrow A = a_1 = \frac{1}{\sqrt{14}} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$B = a_2 - a_1 \frac{a_1^T a_2}{a_1^T a_1}$$

$$= a_2 - \frac{6}{14} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 - 6/14 \\ 1 - 12/14 \\ 1 - 18/14 \end{bmatrix} = \frac{1}{14} \begin{bmatrix} 8 \\ 2 \\ -4 \end{bmatrix} = \frac{1}{7} \begin{bmatrix} 4 \\ 1 \\ -2 \end{bmatrix}$$

$$\textcircled{4} \quad 4 + 4 \quad \lambda_1 \quad \lambda_2 \quad \lambda_3 \quad \lambda_4$$

$$a) \Rightarrow \lambda_i \neq 0 \quad \lambda_i \neq 0 \quad \text{for invertible}$$

$$b) \quad \det(A^{-1}) = \frac{1}{\det(A)} = \frac{1}{\lambda_1 \lambda_2 \lambda_3 \lambda_4}$$

$$c) \quad \det(A + I) = \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + 4$$

⑤ $A \rightarrow \text{tridiag}(1, 1, 1)$

$$D_n = \det A_n$$

a) Use cofactors to show

$$D_n = -D_{n-1} + -D_{n-2}$$

$$\det A_n = 1 \cdot D_{n-1} + (-1)(-D_{n-2})$$

$$D_n = D_{n-1} - D_{n-2}$$

b) Solution for system $\rightarrow$

$$D_1 = 1$$

$$D_2 = 0$$

$$\begin{bmatrix} D_n \\ D_{n-1} \end{bmatrix} = \begin{bmatrix} \\ \end{bmatrix} \begin{bmatrix} D_{n-1} \\ D_{n-2} \end{bmatrix}$$

$$u_n = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} u_{n-1}$$

$$\begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$$

$$(1-\lambda)(-\lambda) + 1 = 0$$

$$\lambda^2 - \lambda + 1 = 0$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{1 \pm \sqrt{1-4}}{2}$$

$$\lambda = \frac{1}{2} (1 \pm \sqrt{3} i) \rightarrow \text{stable solution}$$

$$|\operatorname{Re} \lambda|^2 + |\operatorname{Im} \lambda|^2 = \frac{1}{4} + \frac{3}{4} = 1 //$$

Given

$$\lambda^6 = 1$$

as solutions are $e^{i\pi/3}$, $e^{-i\pi/3}$

$$\boxed{A^6 = I}$$

$$\textcircled{6} \quad A_n = \begin{bmatrix} 0 & 1 & & & \\ 1 & 0 & 2 & & \\ & 2 & 0 & 3 & \\ & & 3 & \ddots & \ddots \\ & & & \ddots & \ddots \end{bmatrix}$$

a) $\Rightarrow$ Find projection matrix onto column space of A_3

$$\Rightarrow P = A(A^T A)^{-1} A^T \quad (A^T = A)$$

$$\Rightarrow A(A^T \cdot A)^{-1} A^T$$

$A \Rightarrow$ not invertible

b) Find λ & x

$$\begin{vmatrix} -\lambda & 1 & 0 \\ 1 & -\lambda & 2 \\ 0 & 2 & -\lambda \end{vmatrix} = -\lambda(\lambda^2 - 4) \\ = -1(-\lambda) \\ = \lambda(1 - \lambda^2 + 4) \\ = \lambda(5 - \lambda^2) = 0 \\ (\lambda^2 - 5)(\lambda) = 0$$

$$\boxed{\lambda = 0, \pm \sqrt{5}}$$

c) Projection onto col-space of A_n

$$A_4 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 2 & 0 \\ 0 & 2 & 0 & 3 \\ 0 & 0 & 3 & 0 \end{bmatrix}$$

$A_4 \rightarrow$ invertible

$$A^T = A$$

$$P = I$$